

Comprehensive Technical Report: Pearls AQI Predictor

A 100% Serverless Multi-Horizon Atmospheric Forecasting & MLOps Intelligence Platform

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Executive Summary

The Pearls AQI Predictor is a fully automated machine learning platform designed to forecast the Air Quality Index (AQI) up to three days in advance for over 195 world capitals.

Instead of relying on traditional "black-box" algorithms, I built a transparent consensus engine grounded in the official 0–500 US Environmental Protection Agency (EPA) AQI standard. To get the most accurate predictions, my system blends three different approaches: a statistical Ridge Regression model, an ensemble Random Forest, and a deep learning PyTorch Multilayer Perceptron (MLP).

Tools used to run this are GitHub Actions for the automated data pipelines, Weights & Biases manages our models and data history, and Vercel powers the frontend and API. The result is a sleek dashboard where users can explore 72-hour forecast curves, interactive maps, and straightforward explanations of exactly what is driving their local air quality.

1 System Architecture

The pipeline is designed in five distinct stages, adhering strictly to a serverless mindset to keep costs near zero and ensure instant scalability.

1. Data Ingestion: Live atmospheric data pulled using the OpenWeather API. **2. Feature Engineering:** The raw data is transformed into useful signals (like 24-hour averages and temporal lags) and stored in our Weights & Biases feature store. **3. Model Training:** Trained Ridge, Random Forest, and Deep Learning models, validate them against historical data, and save the best versions. **4. Pipeline Automation:** GitHub Actions run the show—fetching data hourly and retraining models daily without human intervention. **5. Web Application:** A FastAPI backend feeds Astro-based frontend, delivering interactive Leaflet maps and dynamic charts to the user.

Going serverless means no paying for idle computing power. Data harvesting and model training run on ephemeral GitHub Actions runners, while our REST API scales up instantly on Vercel only when a user requests a forecast.

2 Exploratory Data Analysis

Before building any models, I analyzed over 150,000 hourly atmospheric readings across different global climates. Here is what I learned about how pollution behaves:

- **Pollutants travel together:** Fine dust (PM_{2.5}) and coarse dust (PM₁₀) almost always spike at the same time in urban and industrial areas.

- **The Ozone/Traffic trade-off:** Ground-level Ozone (O_3) and Nitrogen Dioxide (NO_2) have an inverse relationship. Morning traffic spikes NO_2 . Then, afternoon sunlight bakes those emissions, turning them into Ozone.
- **Pollution has a daily rhythm:** At night, cooler air traps pollutants close to the ground, causing air quality to drop. Afternoon solar heating creates upward air currents that clear out the smog.
- **Predictability drops over time:** I found that knowing the current air quality makes predicting the next hour very easy. However, predicting 24, 48, or 72 hours out is much harder. To fix this, I built dedicated models for each time horizon rather than predicting one hour at a time and letting errors snowball.

3 Building the Features

Machine learning models are only as good as the data you feed them. The feature pipeline transforms raw weather data into 19 distinct signals, ensuring the model only learns from past events (preventing data leakage).

- **Look-back Lags:** Fed the models the AQI from 1, 2, 3, 6, 12, 24, and 48 hours ago to establish immediate trends.
- **Rolling Statistics:** Calculated the 24-hour average and peak AQI to give the model a sense of the baseline daily pollution.
- **Time & Season Signals:** Used sine and cosine math to help the model understand the cyclical nature of hours in a day and months in a year.
- **Day of the Week:** This helps the model anticipate the difference between heavy weekday commuter traffic and weekend industrial slowdowns.

Because commercial feature stores can be expensive, used Weights & Biases (W&B) Artifacts as a free, serverless alternative to snapshot and version our datasets.

4 Harvesting Historical Data

To ensure our models could handle any climate, from monsoons to arid deserts, Wrote a script to harvest a full year of hourly data for 18 representative global capitals.

Region	World Capitals Included
South & East Asia	New Delhi, Beijing, Tokyo, Islamabad
Europe	London, Paris, Berlin, Madrid
Middle East & Africa	Cairo, Riyadh, Nairobi, Pretoria
Americas	Washington D.C., Mexico City, Brasília, Santiago
Oceania	Canberra, Wellington

This produced a massive dataset of 151,440 hourly records. Crucially, evaluated models using "out-of-time" testing. Instead of randomly splitting the data, the models were trained on the first 80% of the year and tested them on the final 20% to simulate real-world future forecasting.

5 Our Machine Learning Lineup

Pitted three different AI architectures against each other to see which could best predict air quality 24, 48, and 72 hours out.

- **Ridge Regression:** A reliable statistical model that prevents overfitting by penalizing complex feature weights. It handles the linear trends in weather data beautifully.
- **Random Forest:** An ensemble of 100 decision trees. It doesn't need normalized data and is fantastic at catching sudden, non-linear smog anomalies.
- **PyTorch Deep Learning:** A 4-layer neural network using Batch Normalization and Dropout to find hidden, high-level patterns in the atmosphere.

How They Performed

Model Architecture	Test R^2 (+24h)	Test R^2 (+48h)	Test R^2 (+72h)	Mean R^2	Test RMSE (+24h)	Test MAE (+24h)
Ridge Regression	0.7525	0.6551	0.6097	0.6724	22.69	14.18
Random Forest	0.7399	0.6360	0.6029	0.6596	23.26	14.57
Deep Learning (MLP)	0.7021	0.6294	0.5886	0.6400	24.90	15.78

Interestingly, the simplest model won. **Ridge Regression** provided the most stable and accurate forecasts out to three days. However, the **Random Forest** was better at catching extreme smog events. Because each model has distinct strengths, our final production API averages their predictions together to create a highly reliable "consensus" forecast.

6 Making Sense of the Predictions (SHAP & Health Alerts)

No one trusts a black-box AI. We integrated TreeSHAP into our pipeline to explain exactly why the model is making a specific prediction. The platform features an interactive drawer where users can see which factors (like recent PM_{2.5} levels or the time of day) are driving their local forecast up or down.

Also strictly follow the official US EPA math to calculate the final AQI score from raw pollutant concentrations:

$$I_p = \frac{I_{Hi} - I_{Lo}}{BP_{Hi} - BP_{Lo}}(C_p - BP_{Lo}) + I_{Lo}$$
$$AQI = \max_{p \in \{PM_{2.5}, PM_{10}, NO_2, O_3, CO\}} (I_p)$$

When the predicted AQI crosses into dangerous territory (like "Unhealthy" or "Hazardous"), the dashboard automatically renders prominent health advisories and preventive actions based on EPA guidelines.

7 The Automated CI/CD Engine

I wanted this platform to run itself. Using GitHub Actions, I set up completely autonomous workflows. Every single hour, a script wakes up, fetches the latest global weather data, and updates our feature store. Every day at 02:00 UTC, another script retrains all the models on the newly collected data, checks to ensure accuracy hasn't dropped, and deploys the fresh models to production.

8 The Geospatial Dashboard

I tied all of this backend intelligence together with a fast, modern frontend built on Astro and FastAPI. Users are greeted with a dark-mode interactive Leaflet map pre-loaded with over 195 capitals. Clicking any city instantly pulls up a smooth 72-hour prediction curve powered by Chart.js. Users can toggle between the individual models, view the consensus prediction, and see exactly which pollutant is causing issues on any given day.

9 Conclusion

By combining smart feature engineering, a diverse lineup of models, and aggressive serverless automation, I created a self-sustaining platform that delivers accurate, explainable air quality forecasts to anyone in the world.